

Ionic Mobility within Functionalized Silica Nanopores

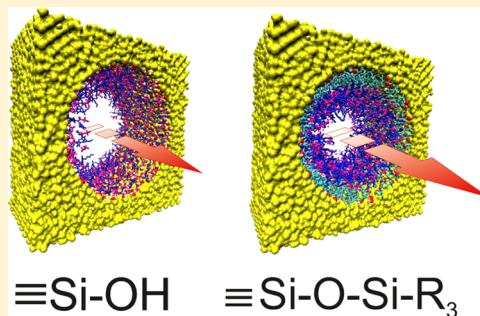
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Supporting Information

ABSTRACT: Molecular dynamics simulations were performed to investigate the structural and dynamical features of an aprotic ionic liquid confined within two types of cylindrical silica pores (hydrophilic and hydrophobic ones) as a function of the pore filling fraction. Analysis of the local density distributions revealed the existence of a dense adsorbed layer in both pores, leading to interfacial ionic liquid densities that resulted between 2 and 3 times larger than bulk. Beyond the characteristics of the surface, it is observed that the nearest-to-the-wall-adsorbed ionic liquid cations accommodate their rings and alkyl chains parallel to the pore wall. Nevertheless, the orientation of the alkyl chain of the more distant cations in the adsorbed layer depends on the functionalization of pore walls, pointing toward the center of the pore for the case of hydrophilic surfaces or toward the pore surface when the wall is covered by hydrophobic moieties. Transport properties were also investigated. The axial translational diffusive dynamics exhibits an overall slowdown upon confinement, being more pronounced in the hydrophilic cavities at low loadings, in agreement with recent experimental results. The ionic conductivity measured in the hydrophilic pores resulted ~50% lower than in the bulk phase. In contrast, within the hydrophobic pores, the conductivity resulted 30% larger than in hydrophilic cavities and showed weak dependence on loading. The contributions to the collective conductivity, arising from single and distinct components, were analyzed and discussed in terms of microscopic correlations and local densities.



1. INTRODUCTION

Ionic liquids (ILs) are fluids composed solely of ions. Owing to their particular physicochemical properties, such as low volatility, low vapor pressure, high thermal stability, and high ionic conductivity,^{1–3} ILs constitute interesting candidates as alternative solvents for the development of new applications in materials science. To cite only a few relevant examples, applications in photochemistry,^{4,5} chromatography,^{3,6–8} separation techniques,^{9,10} catalysis,^{11,12} electrochemistry,¹³ and drug delivery systems^{2,14,15} have been reported in recent years.

The combination of an inorganic porous material filled with an IL gave rise to a novel class of hybrid materials, often called ionogels, with promising applications. The development of such materials is technologically challenging because the properties of the ILs are modified upon confinement.^{16,17} The incorporation of a liquid into a solid host material generally reduces the overall liquid mobility, resulting in smaller diffusion and lower ionic conductivity.^{18,19} Therefore, new strategies must be evaluated in order to minimize these drawbacks. The main scientific challenge in this area consists in the development of new materials that are able to maintain a high conductivity when the IL is confined. Technological applications would require, in addition, the stability of the system properties under high temperatures and pressures, such as in the case of fuel cells.²⁰

The use of different kinds of silica microparticles can help to minimize the mobility reduction in silica-supported ionogel devices.^{21–24} Because the origin of this loss in liquid mobility is

based on IL/silica interactions at the interface, a modification of the pore surface chemistry is therefore a valuable tool in the attempt to reduce the slowdown effect in the IL dynamics.

In a recent work, Garaga et al.²⁵ investigated transport properties of a new type of ionogel, consisting of nanoporous silica microparticles filled with an IL. This material was designed to provide well-connected nanopores, good mechanical properties, and a high surface-to-volume ratio. From Raman spectroscopy and NMR experiments, the authors show that the ionic mobility is significantly enhanced if the silica surface is functionalized with tributylsilyl (TBS) groups. The obtained ionic conductivities resulted in one order of magnitude higher than those observed in the case of untreated silica. The latter observation holds true for both aprotic and protic ILs considered in the experiments, which suggests that the enhancement effect would be independent of the specific identity of the IL. The authors propose a microscopic mechanism as the explanation for this enhancement of the ionic mobility, based on weaker IL/silica interactions.

Similar effects were found upon pore functionalization in the experimental work of Iacob et al.,²¹ in which they investigate the transport properties of 1-hexyl-3-methylimidazolium hexafluorophosphate IL in nanoporous silicon membranes by spectroscopic techniques and pulsed field gradient NMR. A 10-

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fold decrease in the diffusion coefficient of the IL confined in the hydrophilic silica nanopores was found as compared to the bulk value. This drop was explained by the authors in terms of the presence of hydrogen bonding between the IL and the silanol groups. Consistently, the silanization of the porous membranes reversed that trend and resulted in a significant diffusion increment, bringing the diffusion coefficient of the IL in close approach to the bulk value, showing the fundamental role that liquid/surface interactions play in determining the dynamics of nanoconfined systems.

A large body of studies have reported slower dynamics of ILs confined within nanocavities or silica slabs.²⁶ In contrast, theoretical methods have also predicted the opposite trend, that is, a faster dynamics of ILs in nanocavities. Namely, Shi and Sorescu²⁷ have investigated the sorption of CO₂ and H₂ into an IL confined within carbon nanotubes. Their predictions show self-diffusion coefficients that are about 1–2 orders of magnitude larger for the trapped IL than for the corresponding bulk phases. Experimental evidence of dynamical enhancements upon confinement has also been reported. An increment in the diffusivities by more than two orders of magnitude has been measured by Jacob et al.²⁴ using spectroscopic techniques in ILs contained in nanoporous membranes. Neutron scattering experiments performed by Chathoth et al.²⁸ have shown that ILs confined in mesoporous carbon have diffusion coefficients that are comparatively larger in the confined state. Similar diffusion enhancements have been measured when the ILs are confined in carbon nanotubes.^{29,30} Besides, taking into account the specific molecular details of the confining cavity, the increment of ionic mobility in confined phases is attributed to changes in ion packing because of the reduction of the density of the trapped liquid in the pore. In the absence of strong IL/surface wall interactions, the dynamics seems to be controlled not only by thermal but also by density fluctuations.

In the current work, we will present results from molecular dynamics (MD) simulations performed on a model mesoporous system, containing an IL trapped within cylindrical silica nanopores of diameter ~ 8 nm. The chosen IL was the imidazolium-based aprotic 1-methyl-3-hexylimidazolium bis-(trifluoromethanesulfonyl)imide (C₆C₁ImTFSI), similar to one of the two ILs employed in the experiments of Garaga et al.²⁵ Systems with different filling fractions of IL were investigated, within two types of cavities: hydrophilic pores, in which the inner silica surface is covered by polar silanol groups and hydrophobic-like ones, in which 50% of the hydroxyl groups are replaced by TBS chains.

The organization of the paper is as follows: in Section 2, we summarize the details of the model and some technical aspects of the simulation procedure. Section 3 contains the main results of our work, in which we investigate and discuss the structural and dynamical features of the confined IL. Finally, the concluding remarks are presented in Section 4.

2. COMPUTATIONAL DETAILS

Confinement effects on the C₆C₁ImTFSI IL were examined in fully periodic systems comprising cylindrical silica pores containing trapped IL at room temperature. To build up the solid substrate, we followed closely the procedure described in previous publications,^{31–36} so we will briefly describe the scheme here. We started by considering a sample of fused SiO₂, in a simulation box of dimensions $L_x = L_y = 115.4$ Å and $L_z = 57.7$ Å, equilibrated at temperatures close to $T \approx 8000$ K. After immobilizing a central cylindrical section of radius $R_p = 4$

nm along the z -axis, the surrounding portion was allowed to cool down to ambient conditions by multiple rescaling of the atomic velocities. The central portion of the sample was then removed, leaving a cylindrical pore of radius R_p , across the silica slab with dynamical characteristics similar to a solid.

We investigated two types of pores, differing in the chemical functionalization of the internal walls: (i) hydrophilic cavities, in which unsaturated oxygens at the internal walls were hydroxylated, yielding a density of SiOH groups of ~ 4 nm⁻², in reasonable agreement with data obtained from gas adsorption experiments³⁷ and (ii) hydrophobic-like pores, obtained by replacing 50% of the silanol groups by flexible TBS chains. Figure 1 shows typical configurations for the two silica pores investigated. From then on, hydrophilic and hydrophobic-like cavities shall be labeled as “HIC” and “HOC”, respectively.

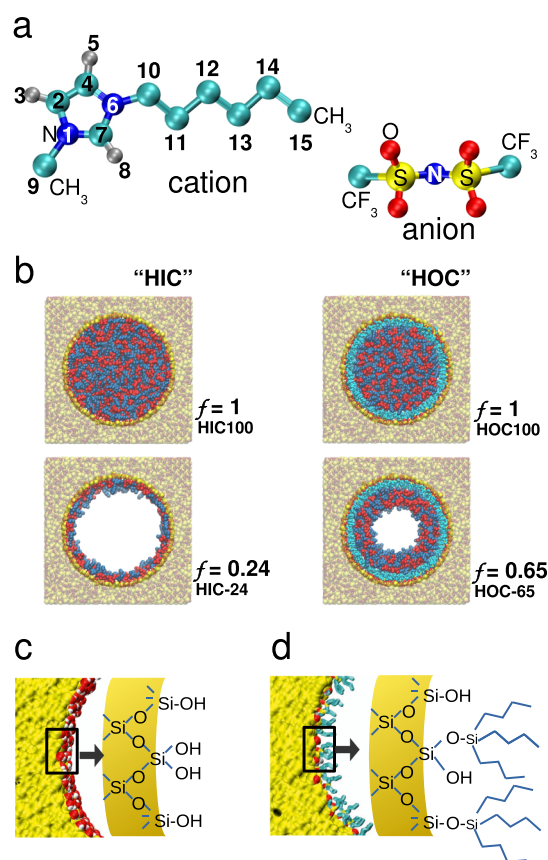


Figure 1. (a) Molecular structure of the IL moieties, [1-hexyl-3-methyl imidazolium bis(trifluoromethane sulfonyl)imide (C₆C₁ImTFSI)], employed in this work. (b) Typical snapshots of selected systems during MD simulations; (c) hydroxylated pore (hydrophilic, HIC) and (d) hydrophobic cavity (HOC), functionalized with TBS groups, along with the schematic illustration of the molecular details at the pore surface.

The preparation stage involved the incorporation of the liquid phase inside the silica pore, comprising N_{ip} ionic pairs of the IL. In the case of full loading, the number of ionic pairs was obtained from a previously equilibrated simulation, in which the lateral faces of the silica block were brought in contact with two adjacent reservoirs containing bulk phases of the IL; this led to a periodic simulation box of linear dimensions $L_z = 160$ Å. During the preliminary simulation, the IL was allowed to

enter the pore from the lateral reservoirs and the system was equilibrated at $T = 300$ K, for about 10 ns. During this period, the length of the simulation box along the z -axis was adjusted to set the density of the IL away from the pore (within the reservoirs region) to that of the bulk at the same temperature. After this preliminary equilibration period, the lateral reservoirs were removed and the linear dimensions of the periodic box along the z -axis were brought back to $L_z = 57.7$ Å. The number of ionic pairs trapped within the pore after this procedure represented the 100% filling fraction ($f = 1$); systems with smaller filling percentages were generated by removing appropriate numbers of ionic pairs. The final numbers of ionic pairs and molecules considered in each simulated system are listed in Table 1.

Table 1. Details of the Simulations^a

pore surface	system label	N_{ip}	N_{SiOH}	N_{TBS}	f (%)
SiOH	HIC100	565	638		100
SiOH	HIC-72	405	638		72
SiOH	HIC-48	270	638		48
SiOH	HIC-24	135	638		24
SiOH/TBS	HOC100	394	319	319	100
SiOH/TBS	HOC-73	287	319	319	73
SiOH/TBS	HOC-65	255	319	319	65

^a N_{ip} , N_{SiOH} , and N_{TBS} correspond to the number of IL ion pairs, SiOH groups, and TBS groups, respectively, present in each of the simulated systems; f is the filling factor, computed as the fraction of molecules within the pore with respect to full loading. HIC and HOC refer to hydrophilic- and hydrophobic-like nanopores, respectively.

At this point, it is worth mentioning that we failed to generate homogeneous systems within HOC pores with filling fractions below $f \approx 0.60$. This was because of the fact that liquid–liquid interactions are stronger than liquid–wall ones in the hydrophobic pores, which leads to collective aggregation of the trapped IL. Below the monolayer threshold, this relative unbalance in the intermolecular interaction strengths produces a heterogeneous distribution of the IL inside the pore, with a structure that strongly depends on the initial state. Therefore, one should study with some care the experimental works that report IL properties measured along the full range of the filling percentage in functionalized pores. We believe that a more realistic picture of the situation at small f would be represented by an ensemble of filled and empty pores that, in average, has an equivalent value of f .

To model the imidazolium-based IL, we adopted the flexible model developed by Zhong et al.,³⁸ which yields reasonable predictions for the structural and dynamical characteristics of ILs, when compared with direct experimental information. The Hamiltonian is based on a united-atom description with CH_2 , CH_3 , and CF_3 groups treated as single units, while all the hydrogens in the imidazolium ring are explicitly considered, allowing a more realistic description of hydrogen bonding in the liquid phase. In addition, the magnitude of the charges on the cation and the anion species were rescaled by 0.8, to compensate the lack of the explicit incorporation of polarization contributions. Intramolecular interactions included stretching, bending, and dihedral contributions. The overall potential energy of the system was decomposed into site–site pairwise interactions combining dispersion (Lennard-Jones) and Coulomb contributions. The inter- and intramolecular potential parameters employed for silica, silanol, and TBS can

be found in previous work;³⁶ partial charges assigned to these atomic sites can be also found in ref 36.

The MD trajectories were generated using the NAMD package³⁹ and corresponded to the NVE microcanonical ensemble. Short-ranged intermolecular forces were cut off at 13 Å, while the particle mesh Ewald method was implemented to handle long-range electrostatic forces. Each system was equilibrated for about ~ 10 ns at $T = 300$ K; after this initial period, configurations were saved every 2 ps along ~ 100 ns runs to collect appropriate statistics.

3. RESULTS

3.1. Structure. We start our analysis by investigating the structural features of the IL along the cylindrical radial direction. This information can be extracted from the local density fluctuations, namely

$$\rho(r) = \frac{1}{2\pi r L_z} \sum_i \langle \delta(r_i - r) \rangle \quad (1)$$

where r_i corresponds to the distance between the i -th site and the pore axis. In Figure 2, we present results for the radial

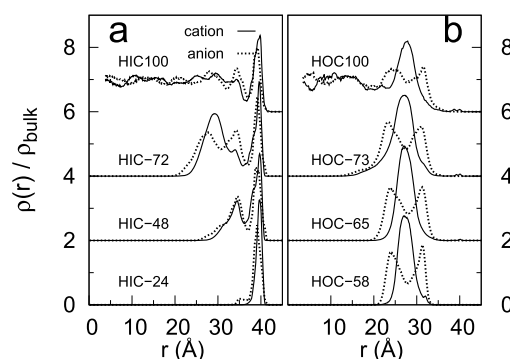


Figure 2. Normalized local density of cation and anion species as a function of the radial coordinate for (a) the hydroxylated and (b) functionalized silica pores at different filling factors. The center of the imidazole ring and the central nitrogen site were taken as molecular centers of cations and anions, respectively. The curves on panel (a) corresponding to HIC100, HIC-72, and HIC-48 systems have been shifted vertically by +6, +4, and +2, respectively, for better visualization. Similar vertical shifts were applied to the curves corresponding to HOC100, HOC-73, and HOC-65 systems, shown in panel (b).

density profiles in HIC and HOC pores, normalized by the bulk density ($\rho_{\text{bulk}} = 0.00365 \text{ Å}^{-3}$), at different filling fractions. In the fully filled hydroxylated cavity (HIC100), a bulklike structureless density profile, $\rho(r)/\rho_{\text{bulk}} \approx 1$, is observed at the central region, at $r \lesssim 25$ Å. As we approach the silica walls, important fluctuations are perceived. Most notably is a local maximum of height ~ 2 is observed at $r \approx 39$ Å for both species, with the cation ring located slightly closer to the wall than the anion. Hydroxylation also promotes a large extent of the liquid structure toward the interior of the pore, as revealed by the subsequent local maxima—of smaller amplitude—in the profiles at intermediate radii, $25 \text{ Å} \lesssim r \lesssim 39 \text{ Å}$. The density profiles in the fully filled HOC pore (HOC100), shown in Figure 2b, differ considerably from HIC ones. First, the presence of the bulky and hydrophobic TBS chains at the silica surface promotes a broader cation peak of height $\rho(r)/\rho_{\text{bulk}} \approx 2$ at a smaller radius, $r \approx 28$ Å, flanked by two anion layers of

roughly half the cation amplitude. At smaller filling fractions, the density profiles near the silica preserve a similar layering structure to those for $f = 1$, suggesting that in the HOC case, a $\sim 30\%$ reduction in the pore loading is sufficient to stabilize an adsorbed monolayer at the pore wall. Contrasting, in HIC pores, in addition to the strongly adsorbed layer at $r \approx 39$ Å, the gradual decrease of the filling fraction shows marked modifications in the structure of the second layer at $r \approx 20$ – 35 Å.

Additional structural details appear in Figure 3, where the distributions of some relevant sites are displayed along the

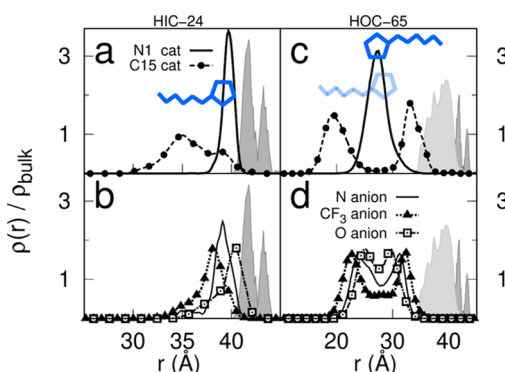


Figure 3. Details of the local density distribution for a set of selected atoms: N1 and C15 sites of the cation, and N, CF₃, and O sites of the anion. Top panels correspond to cation sites and bottom ones correspond to the anion ones. Left and right panels correspond to HIC-24 and HOC-65 systems, respectively. The shaded dark and light gray curves represent the density of interfacial SiOH and TBS groups, respectively.

radial coordinate. For cation species, the selected sites are the nitrogen (N1) and the terminal CH₃ group (C15). For anions, the central nitrogen, the two CF₃ groups, and the four oxygen sites were chosen. For simplicity purposes, we will restrict the present description to the systems with the lowest filling fractions: HIC-24 and HOC-65. The left panels of Figure 3 correspond to the HIC-24 system and show that near the silica surface, the cation site N1 lies in intimate contact with the SiOH groups (see the narrow and intense peak at $r \approx 40$ Å). The density distribution associated to the terminal carbon in the hexyl chain of the cation (C15) exhibits a broad peak centered at $r \approx 35$ Å, associated to the interfacial cations that have their hexyl tail oriented perpendicularly to the pore walls and pointing toward the center of the cavity. A small shoulder at $r \approx 40$ Å is also perceived in $\rho_{C15}(r)$, suggesting that a minor fraction of the hexyl cation tails is oriented parallel to the surface. Anion species in the HIC-24 system (Figure 3b), in turn, orient their oxygens toward the silica surface, most likely forming hydrogen bonds with the SiOH groups, and accommodate their CF₃ groups toward the interior of the pore. Site distributions for the HOC-65 system (displayed in the right panels of Figure 3) reveal that in the functionalized pore, the cations in close proximity with the pore walls are flipped around as compared to the previous scenario. While in HIC-24 a large fraction of cations in the adsorbed layer exhibits their alkyl chains toward the interior, in HOC-65, the majority of the cations accommodate their hydrocarbon chains toward the wall of the pore. In HOC, the density peak closest to the pore surface corresponds to the terminal C15 site of the cation in the HOC system, at $r \approx 35$ Å, followed by an inner

peak at $r \approx 28$ Å—twice as intense as the former—corresponding to the N1 cation sites. This ordering of the density peaks suggests that the adsorbed cations are arranged perpendicularly to the surface. More specifically, about 50% of cation molecules have their hexyl chains pointing toward the pore surface while the rest shows the opposite orientation, as denoted by the second peak of $\rho_{C15}(r)$, centered at $r \approx 20$ Å. The cation peak corresponding to the N1 site in the HOC-65 system is flanked by two layers of anion species, which orient their oxygens in close contact with the cation ring and their CF₃ groups pointing outward, as can be seen in Figure 3d.

To gain a more detailed insight into the orientation of wall-solvation states, we examined local orientational correlations of the confined IL cation species, in terms of the angles

$$\cos \theta_r = \hat{n} \cdot \hat{u}_{\text{ring}} \quad \text{and} \quad \cos \theta_a = \hat{n} \cdot \hat{u}_{\text{alkyl}} \quad (2)$$

In the previous equations, $\hat{n}$ is a unit vector perpendicular to the silica surface; whereas $\hat{u}_{\text{ring}}$ represents a unit vector perpendicular to the imidazolium ring and $\hat{u}_{\text{alkyl}}$ is parallel to the alkyl chain direction, $[r(\text{C15}) - r(\text{C10})]$ (see scheme on

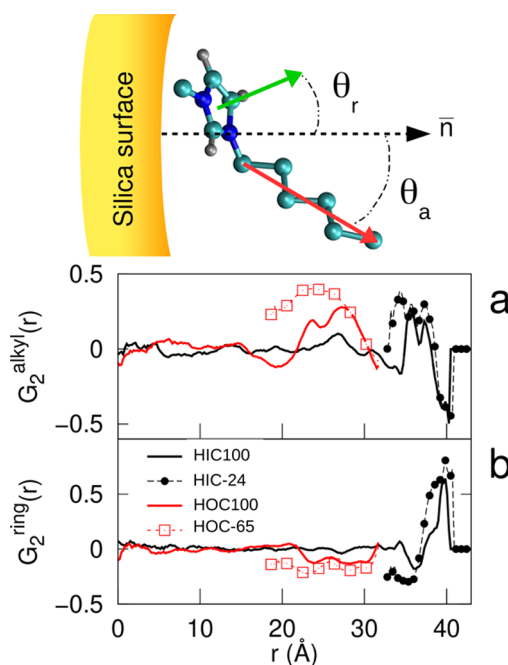


Figure 4. Orientational correlation functions for the IL cation confined within HIC (black lines) and HOC pores (red lines), with different filling fractions.

top of Figure 4). We focused the attention on distribution functions of the type

$$G_2^\alpha(r) = \left\langle \frac{1}{n_r} \sum_i P_2(\cos \theta_{i,\alpha}) \right\rangle \quad (3)$$

where $P_2(x) = (3x^2 - 1)/2$ is the second-rank Legendre polynomial and the sum is restricted to IL cations which lie in a $\Delta r = 0.5$ Å interval, centered at r , containing n_r molecules and $\alpha = \text{alkyl}$ or ring . Note that $P_2(\cos \theta_a) \approx 1$ and ~ -0.5 for alkyl chains are perpendicular and parallel to the silica surface, respectively. On the other hand, in the case of the imidazolium orientations, $P_2(\cos \theta_r) \approx 1$ and ~ -0.5 for cation rings are parallel and perpendicular to the silica surface.

Figure 4 shows the distribution functions $G_2^{\text{alkyl}}(r)$ and $G_2^{\text{ring}}(r)$ for the IL cation confined within the HIC and HOC pores for the fully filled condition and the smallest filling fraction considered in each case. The curves displayed in Figure 4 confirm that in the vicinity of the HIC silica surface (black lines), within the first layer in close contact with the wall ($38 \text{ \AA} \lesssim r \lesssim 41 \text{ \AA}$), cation rings and alkyl chains are preferentially adsorbed parallel to the silica surface. At slightly smaller radii within the same layer ($35 \text{ \AA} \lesssim r \lesssim 38 \text{ \AA}$), the cation alkyl chains and imidazole rings arrange mainly perpendicular to the silica surface. Moreover, with the help of Figure 3, it can be concluded that the alkyl chains located in that radial range point toward the inside of the pore. The black lines with circles, corresponding to the HIC-24 system, show that similar conclusions may be reached regarding the alkyl and ring orientations. Further on, closer to the center of the pore ($r \lesssim 30 \text{ \AA}$), both orientational correlations exhibit values around zero, corresponding to bulklike properties with no clear orientational preference.

In contrast, near the functionalized silica surface containing TBS groups (red lines), we observe that $G_2^{\text{alkyl}}(r) > 0$ and $G_2^{\text{ring}}(r) > 0$ within the first adsorbed layer. This indicates that cation rings and alkyl chains are most likely arranged perpendicular to the pore surface. Again, Figure 3 tells us that the cations closest to the wall point their alkyl chains toward outside. No ordering preference is observed in the $r \lesssim 15 \text{ \AA}$ bulklike region.

The above observations, based on the orientational correlations, confirm and complement the picture obtained from the site density profiles shown in Figure 3.

3.2. Dynamical Properties. **3.2.1. Diffusion.** The translational dynamics of the trapped IL can be evaluated by computing self-diffusion coefficients from the classical Einstein relation

$$D_z = \lim_{t \rightarrow \infty} \frac{\mathcal{R}^2(t)}{2t} = \lim_{t \rightarrow \infty} \frac{\langle [z_i(t) - z_i(0)]^2 \rangle}{2t} \quad (4)$$

where $z_i(t)$ represents the center-of-mass position of molecule i at time t projected along the z -axis, and $\mathcal{R}^2(t)$ is the ensemble-averaged mean squared displacement (MSD). $\mathcal{R}^2(t)$ curves can be found in the Supporting Information section, while the complete set of self-diffusion coefficients is listed in Table 2. Owing to the slow dynamics of the ILs, the linear regime in the MSDs is reached after characteristic times of the order of ~ 4 – 5 ns. A numerical comparison between the bulk diffusion coefficients computed in this work and experimental data⁴⁰

Table 2. Diffusion Coefficients for Cation and Anion Species (in $10^{-7} \text{ cm}^2 \text{ s}^{-1}$) Normalized Values ($\tilde{D} = D/D_{\text{bulk}}$), along the Axial Direction

system	cation		anion	
	D_z	$\tilde{D}_z$	D_z	$\tilde{D}_z$
bulk	0.879	1	0.722	1
HIC100	0.423	0.48	0.308	0.43
HIC-72	0.305	0.35	0.217	0.30
HIC-48	0.122	0.14	0.090	0.12
HIC-24	0.039	0.04	0.019	0.03
HOC100	0.678	0.77	0.585	0.82
HOC-73	0.711	0.80	0.613	0.84
HOC-65	0.694	0.78	0.624	0.85

(i.e., $1.84 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ for cation species and $1.68 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ for anion ones) shows that the force field employed here predicts values that are nearly a factor of two smaller than the experiments. Nevertheless, this underestimation is in agreement with previous numerical results obtained using the same force field. Garaga et al.²⁵ have reported a 50% increment in the cation diffusion coefficient in passing from HIC100- to HOC100-like systems; our simulations predict a similar 60% increment, indicating that although the absolute values of the diffusion are underestimated, our results accurately predict the relative change in IL diffusion upon pore functionalization.

We also found that the dynamics of the confined cations is always faster than that of the anions; cations diffuse ~ 11 – 15% faster than anions in bulk phases and in hydrophobic cavities; this difference increases up to ~ 35 – 40% in the hydrophilic pores, except at the lowest loading, where the diffusion of cations exceeds that of anions by a factor of two. This difference between cation and anion diffusion is typical of [Bmim]-based ILs^{26,41} and has been rationalized in terms of two features: (i) the flexible and mobile alkyl chain in the cation, in addition to the planar structure of the imidazole ring that facilitates its faster dynamics. (ii) On the other hand, the homogeneous electrostatic field of the cations would also contribute to a faster dynamics, contrasting to the more polarized field of the anion.⁴² Within the hydroxylated pores, the normalized diffusion coefficients fall below half of the bulk values, exhibiting progressive retardations as the filling fraction is reduced. At the monolayer limit (HIC-24 system), the diffusion resulted ~ 10 – 15 times slower than in fully filled pores.

The functionalization of the inner surface of the silica pore with TBS hydrophobic groups produces well-differentiated dynamics. First, the axial self-diffusion at $f = 1$ is much closer to the bulk value than in the HIC pore, reaching values that are nearly 80% of the bulk diffusion, compared to just 40% in the HIC case. Second, the variations of the diffusion coefficient with the filling fraction are much milder in HOC systems than in HIC ones. The slower mobility in the latter systems can be attributed to the strong liquid/wall surface interactions and consequently to the formation of a dynamic adsorption layer at the pore interface. The silanization of the silica walls significantly reduces this effect and leads to an enhancement of the overall dynamics.

Owing to the coexistence of a variety of dynamical modes arising from surface bound and bulklike IL molecules, we found it instructive to define local diffusion coefficients expressed in terms of

$$D_z(r) = \frac{\mathcal{R}^2(r; t_2) - \mathcal{R}^2(r; t_1)}{2(t_2 - t_1)} \quad (5)$$

where the averages $\mathcal{R}^2(r; t)$ are accumulated along cylindrical, concentric layers of thickness Δr , centered at r . The values of t_1 and t_2 were fixed at 3 and 5 ns, respectively, on the assumption that diffusional regimes are reasonably reached after 2 ns. On the other hand, t_2 should be shorter than the residence time of the IL molecules within the considered bin. Because IL molecules were assigned to a particular layer depending only upon their initial positions $\mathbf{r}_i(0)$, the physical meaning of the local diffusion coefficient that we compute illustrates how fast is the solvent leaving a given region of space.

The results of normalized diffusion coefficients, computed along the axial direction, are displayed in Figure 5. The plot shows the total diffusion coefficients, calculated as $D_z = 2D_z^{\text{cat}}D_z^{\text{an}}/(D_z^{\text{cat}} + D_z^{\text{an}})$.

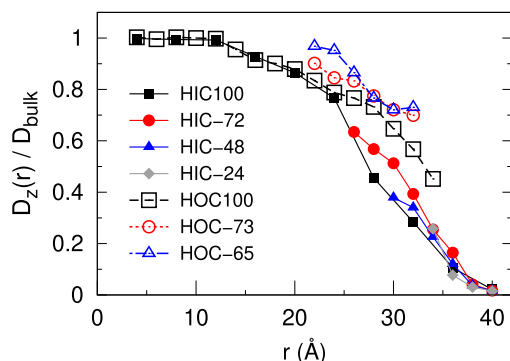


Figure 5. Local self-diffusion coefficients of the IL confined within the silica pores, as a function of the distance to the center of the pore.

The reduction in the dimensionality of available space at the interface because of the presence of the pore walls promotes a general decline in the rate of IL mobility near the inner pore surface and causes the diffusion to deviate from the overall average. An additional reduction in IL mobility arises from the combined effects of the pore wall roughness and IL structuring by means of hydrogen bonding, in the case of the hydroxylated cavities. This effect has been observed in many MD simulations involving aqueous^{43–48} and nonaqueous solvents.^{49,50}

More specifically, the diffusion coefficients of trapped IL within the HIC100 pore exhibit a sharp drop as one approaches the pore walls covered with the polar groups, reaching values at the adsorbed layer ($r > 35$ Å) that are one order of magnitude smaller than those found at the center of the pore. The effects are less marked in the functionalized HOC100 system, in which the reduction in the IL mobility at the interface is of the order of 50%. The results suggest that the average diffusion coefficients computed from eq 4 contain ponderated contributions from surface bound and bulklike molecules. As the filling fraction decreases, the relative population of the faster bulklike particles decreases and that of surface bound ones increases, leading to an overall slower diffusive dynamics. The magnitude of the average self-diffusion coefficient then decreases with the increment of the surface-to-volume population ratio.

3.2.2. Electrical Conductivity. Within the Green–Kubo linear response formulation,⁵¹ ionic conductivities can be calculated in terms of the microscopic charge current time autocorrelation function

$$\sigma = \frac{1}{k_B TV} \int_0^\infty \langle J^\parallel(t) \cdot J^\parallel(0) \rangle dt \quad (6)$$

$$J^\parallel(t) = \sum_{i=1}^N q_i \dot{z}_i(t) \quad (7)$$

In the previous equations, k_B is the Boltzmann constant, V denotes the pore volume, $J^\parallel(t)$ is the parallel charge current along the axial direction, q_i is the electrical charge of ion i , and $N = N_{\text{cat}} + N_{\text{an}}$ is the total number of ions. By integrating by

parts and invoking time symmetry, eq 6 can be transformed into the equivalent Einstein–Helfand relation^{52–55}

$$\sigma = \lim_{t \rightarrow \infty} \frac{1}{2tk_B TV} \times \left\langle \sum_{i,j}^N q_i [z_i(t) - z_i(0)] q_j [z_j(t) - z_j(0)] \right\rangle \quad (8)$$

Note that the angular bracket in eq 8 contains a collective quantity that includes single ($i = j$) and distinct ($i \neq j$) contributions. As such, its calculation is likely to be subjected to larger errors compared to the case of single-particle properties. By analogy with eq 4, we will call the ensemble average between angular brackets in eq 8 as “mean squared charge displacement” (MSCD).

By considering only the self-terms in eq 8, the Nernst–Einstein (NE) approximation for the conductivity is obtained, namely

$$\begin{aligned} \sigma_{\text{NE}} &= \lim_{t \rightarrow \infty} \frac{1}{2tk_B TV} \left\langle \sum_{i=1}^N q_i^2 [z_i(t) - z_i(0)]^2 \right\rangle \\ &= \frac{1}{k_B TV} [N_{\text{cat}} q_{\text{cat}}^2 D_z^{\text{cat}} + N_{\text{an}} q_{\text{an}}^2 D_z^{\text{an}}] \end{aligned} \quad (9)$$

where D_z^{cat} and D_z^{an} are the cation and anion self-diffusion coefficients, defined in eq 4. The NE approximation provides a realistic value for the conductivity in dilute electrolyte or molten salt systems; however, cross correlations in ILs cannot be neglected. Anticorrelated motions of ions of the same charge or correlated ones of ions of opposite charge lead to substantial modifications from the NE estimation. The degree of uncorrelated motion is typically measured in terms of $\alpha = \sigma/\sigma_{\text{NE}}$, the ratio of the collective (total) conductivity to the one due only to self-contributions.

In the present case, our results were obtained using the total porous volume, $V = \pi R_p^2 L_z$, which does not necessarily correspond to the volume occupied by the IL. These two volumes may be equal for fully loaded pores; however, the effective volume occupied by the IL may differ considerably when the pore is partially loaded and the IL is not homogeneously distributed or when the functionalization of the inner pore surface requires the definition of an *effective* pore radius, smaller than R_p . The characteristics of the density profiles displayed in Figure 2 led us to define minimum and maximum radii, R_{min} and R_{max} , taking into account the positions over which $\rho(r)$ has non-negligible values. Results for rescaled conductivities, $\tilde{\sigma} = \sigma(V/\tilde{V})$, with $\tilde{V} = \pi(R_{\text{max}}^2 - R_{\text{min}}^2)L_z$, are shown in columns 2 to 6 of Table 3.

At first glance, we observe that the collective conductivity, $\tilde{\sigma}$, is always smaller than $\tilde{\sigma}_{\text{NE}}$, indicating negative cross

Table 3. Ionic Conductivity of Bulk and Confined IL^a

System	R_{min}	R_{max}	$V/\tilde{V}$	$\tilde{\sigma}_{\text{NE}}$	$\tilde{\sigma}$	α
bulk				1.153	0.847	0.74
HIC100	0	40	1.00	0.568	0.418	0.73
HIC-72	19	40	1.29	0.397	0.308	0.77
HIC-48	26	40	1.73	0.142	0.057	0.40
HIC-24	34	40	3.60	0.024	0.014	0.58
HOC100	0	34	1.38	0.794	0.544	0.68
HOC-73	13	34	1.62	0.848	0.567	0.67
HOC-65	21	34	2.24	1.055	0.572	0.54

^aUnits of conductivities ($\tilde{\sigma}_{\text{NE}}$ and $\tilde{\sigma}$) are mS cm^{−1}; radii are expressed in Å; $\alpha = \sigma/\sigma_{\text{NE}}$.

contributions. This is in agreement with previous experimental and theoretical works.^{55–57} The experimental bulk values for the ionic conductivity reported at $T = 300$ K are close to $\sigma \approx 2$ mS/cm (refs 25 and 58). Our result for the bulk phase, $\sigma = 0.847$ mS/cm, underestimates the IL conductivity by a factor of ~ 2.5 , a value which is consistent with the slower diffusive dynamics predicted by the force field employed in the MD simulations. As discussed by Chen et al. in ref 54, in the calculation of σ from simulations performed with scaled partial charges, one should choose whether q_i in eq 8 is $\pm 0.80 e$ or instead $q_i = \pm 1 e$. The ionic conductivities reported here were computed using scaled charges; to transform them into unscaled ones, σ should be multiplied by a factor $1/0.80^2 = 1.56$. Although debatable, this correction would bring the computed bulk conductivity into closer agreement with the experimental data.

In the HIC pores, $\tilde{\sigma}_{\text{NE}}$ becomes gradually smaller as the filling fraction is reduced, in agreement with the trends observed in the diffusion coefficients with decreasing f . The collective conductivity reflects similar variations with f , being 30 times smaller in HIC-24 than in the fully loaded HIC100 pore. In contrast, in hydrophobic pores, ionic conductivities remain practically constant as a function of loading, exhibiting minimal fluctuations with f around the value obtained in the HOC100 case, $\tilde{\sigma} \approx 0.5$ mS cm⁻¹. This observation indicates that confinement effects on the conductivity in functionalized pores are much milder than in HIC ones. A strong dependence of the conductivity on the pore filling fraction in hydrophilic cavities and a much weaker dependence in functionalized pores were also observed in the experimental study of Garaga et al.²⁵

The comparison of the experimental conductivities reported by those authors in fully filled hydroxylated and functionalized pores shows an enhancement in the molar conductivity of one order of magnitude upon surface functionalization with TBS groups. In our case, the comparison of the conductivities at 100% filling fraction obtained in the two pores yields a more moderate increment, close to 30%. Although the absolute values of the ionic conductivities predicted by our simulations are underestimated with respect to the experimental values, we found good agreement in the difference in conductivity between the bulk phase and the trapped one within the functionalized pore. Namely, both methods, simulations and experiments, agree in predicting relative values of $\sigma_{\text{HOC}}/\sigma_{\text{bulk}} \approx 0.6$.

A detailed interpretation of the changes observed in the collective conductivity in the different confined systems in terms of microscopic features is not straight forward because it involves the interplay of cross correlations between like and unlike species involving bulklike and surface-adsorbed molecules of well-differentiated dynamics. With regard of eq 8, it is also possible to evaluate *local* ionic conductivities,⁵⁹ $\sigma(r)$, along the radial direction, in a similar fashion as performed when dealing with the local diffusion coefficients in eq 5. Therefore, the collective ionic conductivity has been computed in concentric, cylindrical layers of thickness $\Delta r = 2$ Å, centered at r . The results are displayed in Figure 6, along with the corresponding local density of cations and anions.

We notice that the conductivity is nonuniform within the silica cavity. Namely, within the $0 \lesssim r \lesssim 20$ Å inner region of the pore, the ionic conductivity is very close to that of the bulk liquid phase, displayed with a dashed horizontal line in Figure 6. Within the interfacial region, corresponding to more external locations, the ionic conductivity drops down to $\sim 50\%$ of the

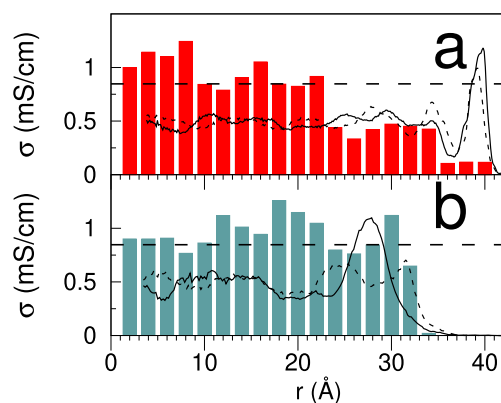


Figure 6. Local conductivities computed for (a) HIC100 and (b) HOC100 systems, as a function of the radial coordinate. The horizontal dashed line corresponds to the ionic conductivity of the bulk phase; solid and dotted profiles correspond to the density distribution of cation and anion species, respectively, in arbitrary units.

bulk value in the HIC case. In contrast, in the HOC pore, the conductivity remains close to the bulk value up to distances of $r \approx 32$ Å. In the HIC pore, the ionic conductivity is roughly one order of magnitude smaller than in bulk within the adsorbed layer, at $r \gtrsim 35$ Å, where the IL molecules are bound to the OH groups of the hydroxylated walls. Although the *average* ionic conductivity is reduced in the presence of silica walls, the computation of *local* conductivities provides a microscopic perspective of the spatial variations in the radial direction.

3.2.3. Contributions to the Collective Conductivity. Previously, we mentioned that the total conductivity of an IL is a result of a complex interplay of ion self-diffusion and correlated ionic motion. In this section, we qualitatively analyze the contributions to the collective conductivity, in order to elucidate the origins of the reduction of σ with respect to σ_{NE} , reported in the previous section. The collective conductivity in eq 8 can be explicitly expressed as the sum of single (s) and distinct (d) contributions

$$\sigma = \sigma_{\text{cat}}^s + \sigma_{\text{an}}^s + \sigma_{\text{cat,cat}}^d + \sigma_{\text{an,an}}^d + 2\sigma_{\text{cat,an}}^d \quad (10)$$

where the subindices “cat” and “an” denote different species and σ_r^s and $\sigma_{r\delta}^d$ are given by

$$\sigma_r^s = \lim_{t \rightarrow \infty} \frac{1}{2tk_{\text{B}}TV} \times \left\langle \sum_i^{N_r} (q_i^r)^2 [z_i^r(t) - z_i^r(0)]^2 \right\rangle \quad (11)$$

and

$$\sigma_{r\delta}^d = \lim_{t \rightarrow \infty} \frac{1}{2tk_{\text{B}}TV} \times \left\langle \sum_i^{N_r} \sum_{j \neq i}^{N_\delta} q_i^r q_j^\delta [z_i^r(t) - z_i^r(0)][z_j^\delta(t) - z_j^\delta(0)] \right\rangle \quad (12)$$

Note that the first two terms of eq 10 account for the NE approximation for the conductivity.

On the basis of eqs 10–12, we decomposed the total MSCD into self and distinct terms, and plot the resulting contributions in Figure 7. On top of the MSCD plots, we also show the fractional contributions of single and cation/anion distinct terms to the total conductivity. By definition, the single components of MSCD are positive and have positive slope; in contrast, distinct terms have no definite sign. We observe that

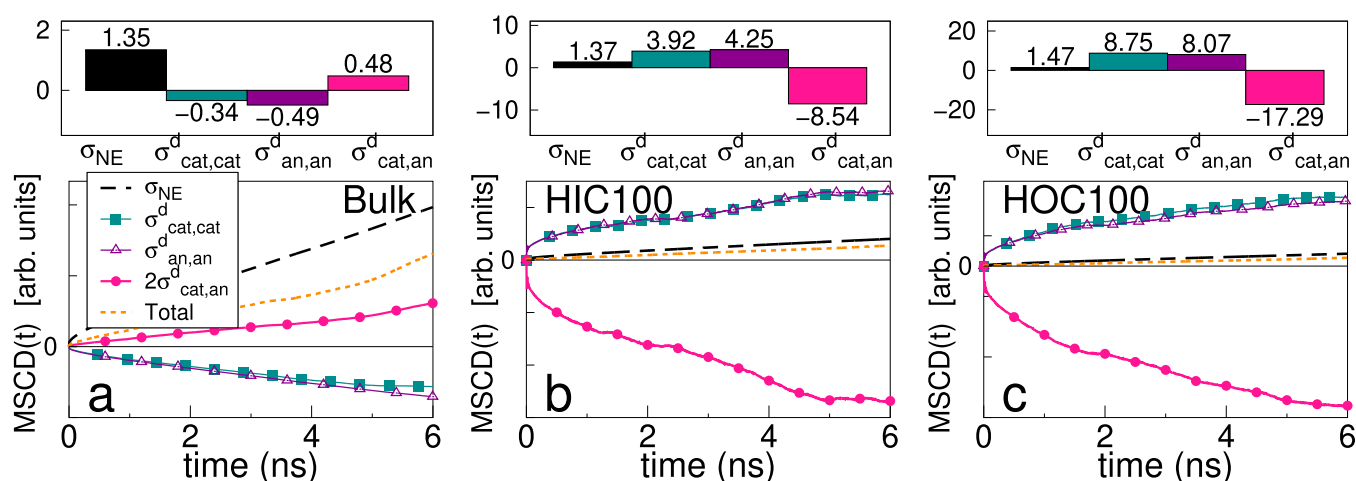


Figure 7. MSCDs and their components from single and distinct terms for the IL in the bulk phase (a) and confined within the hydrophilic (b) and hydrophobic (c) silica pores. The upper panels show the fractional contribution of self and distinct conductivity terms. The plotted values correspond to the ratio of $\sigma_{NE} = (\sigma_{cat}^s + \sigma_{an}^s)$, $\sigma_{cat,cat}^d$, $\sigma_{an,an}^d$, and $\sigma_{cat,an}^d$ to the total conductivity.

in the bulk, the MSCDs of $\sigma_{cat,cat}^d$ and $\sigma_{an,an}^d$ are negative and have negative slope, indicating that the reduction of the collective conductivity with respect to σ_{NE} is mainly because of the overall anticorrelated motion of ions of the same charge. In contrast, the MScD of the $\sigma_{cat,an}^d$ component is positive and enhances the conductivity. However, because the MScD of $\sigma_{cat,an}^d$ contains the product of unlike charges, the correlations between displacements of ions of opposite charge are also anticorrelated in the bulk phase. Similar features have been observed in previous works when decomposing the ionic conductivity of bulk imidazole-based ILs.^{56,60}

An interesting change is observed when one analyzes the components of the MScD for the ILs under confinement; all the distinct terms exhibit signs opposite to those observed in the bulk phase. The magnitude of the relative weight of these terms with respect to the total MScD also exhibits dramatic changes; in HIC100 and HOC100, the MSCDs of the distinct terms $\sigma_{cat,cat}^d$ and $\sigma_{an,an}^d$ are positive and with a positive slope, corresponding to correlated motions of ions of the same charge that enhance the conductivity. By contrast, the negative sign and slope of the MScD of $\sigma_{cat,an}^d$ can be associated to the correlated dynamics of ions of opposite charge that reduces the total conductivity with respect to the ideal NE case. At this point, it is worth mentioning that special care must be taken when interpreting the behaviour of distinct conductivity terms, as these quantities are not fully independent but constrained by momentum conservation conditions.^{56,61,62} As a consequence, the distinct conductivities are reference-frame dependent. For simplicity, we have considered in all cases the laboratory reference frame, in which the center-of-mass momentum of the system is zero. Without further discussions about this technical detail, we shall now focus on the physical interpretation of our findings.

Positive contributions to the total conductivity arising from distinct terms that involve ions of same charge and negative distinct components from ions of opposite charge have been already observed in previous works on IL aqueous solutions under nanoconfinement.¹⁸ Similar features were reported in recent studies of IL solutions of the $[BMIM^+][BF_4^-]$ IL in a low-dielectric solvent (dichloroethane) at small concentrations.⁶⁰ The cross dynamical correlations are largely determined by the characteristics of the local densities. In

this sense, we believe that the layering structure near the interface would enhance cooperative effects between ions within the layer. In fact, the relative weight of distinct conductivities is roughly 2 times larger in the HOC100 pore than in the HIC100 one. This difference could be rationalized by recalling that in HOC100, cation and anion species are nearly segregated within adjacent coaxial sections close to the pore walls, whereas in HIC100, cation and anion density peaks are mostly overlapped within the adsorbed layer, and therefore, a greater degree of cancellations is likely to occur in this case. More specifically, in the present context, the positive contribution of the distinct terms $\sigma_{cat,cat}^d$ and $\sigma_{an,an}^d$ to the conductivity in HIC100 and HOC100 systems may be cast in terms of like-ion correlations that would be facilitated by the counterions, when the ions are paired; within this physical picture, a cation or anion would drag along its associated counterion because of its motion, which, in turn, pulls along ions of the same charge as that of the initial one. On the other hand, the very large magnitude and negative sign of the cation/anion distinct term $\sigma_{cat,an}^d$ might be evidence of ion clustering or ion-pairing aggregation.

In order to elucidate the existence of short- or long-lived ionic pairs within the studied systems and evaluate whether their motion as neutral pairs is responsible for the reduction of the conductivity as compared to σ_{NE} , we will analyze ion-pair correlation functions and ion-pair lifetimes in the next section.

3.2.4. Ion-Pair Association. Microscopic inspection of ion aggregation can be evaluated by computing correlation functions of the type⁶³

$$P(t) = \sum_{ij} \langle \eta_{ij}(t) \eta_{ij}(0) \rangle \quad (13)$$

with $\eta_{ij}(t) = 1$ if the cation/anion pair ij is associated at time t and zero otherwise. We have considered the so-called continuous approach for the time correlation function. In this case, each state η_{ij} is allowed to make just one transition from unity to zero when the pair is first observed to break, but is not allowed to return to unity if the same pair associates subsequently at later times.⁶⁴ In this way, this analysis provides an estimation of the period of time during which a ion pair remains continuously associated. Based on the radial pair correlation function (displayed in the [Supporting Information](#)

section), cations and anions are considered as an associated ionic pair if they are separated by a distance closer than $d = 7.5$ Å.

The results for $P(t)$ are shown in Figure 8. The estimates for the characteristic aggregation times, τ_{ip} , are calculated from,

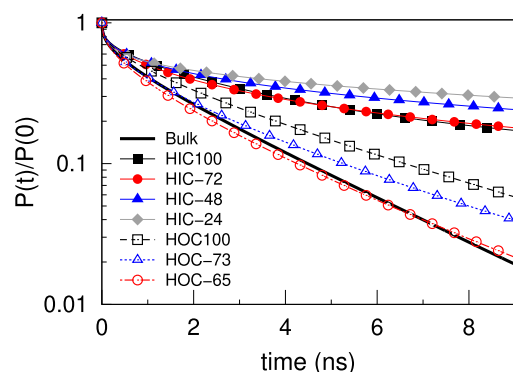


Figure 8. Time correlation function $P(t)$ of cation/anion aggregation for the IL in different environments, as indicated.

$$\tau_{ip} = \int_0^\infty \frac{P(t)}{P(0)} dt \quad (14)$$

that is, the time integrals of the normalized time correlation functions, fitted to single exponentials at long times. The best-fitting parameters and the obtained aggregation times are listed in Table 4, along with the percentages of associated pairs in each system, $a_{ip} = P(0)/N_{ip}^2$.

Table 4. Best-Fitting Parameters for the Correlation Function $P(t)/P(0) \approx A_0 \exp(-t/\tau)$, Aggregation Times τ_{ip} , and Percentage of Associated Pairs, a_{ip}

system	A_0	τ/ns	τ_{ip}/ns	$a_{ip} \times 100$
bulk	0.525	2.71	1.58	0.3
HIC100	0.433	9.40	4.46	0.6
HIC-72	0.405	10.72	4.72	0.8
HIC-48	0.449	14.18	6.68	1.1
HIC-24	0.471	18.50	8.97	1.5
HOC100	0.485	4.19	2.20	0.9
HOC-73	0.397	3.83	1.84	1.2
HOC-65	0.406	3.02	1.50	1.3

The aggregation time obtained for the bulk IL was $\tau_{ip} = 1.58$ ns. Similar aggregation times were found in the hydrophobic HOC pores, lying in the range 1.5–2.2 ns and exhibiting a weak dependence on the filling factor. In contrast, a substantial increase in τ_{ip} from ~ 4 ns up to ~ 9 ns is observed within HIC pores when decreasing the filling fraction. As we shall see below, this is because of the slower translational dynamics, which makes ion pairs to remain together over longer times. This result is consistent with experimental findings in similar ILs confined within the disordered silica matrix.⁶⁵ We also note that the average fraction of associated pairs a_{ip} , although small, increases up to five times with the decrease of f , as compared to the bulk phase.

If the association into long-lived ion pairs that move together as neutral units was responsible for the large (negative) magnitude of cation/anion distinct contributions to the conductivity in the confined systems, these ion pairs

would have to travel together as a pair—without breaking their contact—over sizable distances. This can be studied from the MD trajectories, from a combined distribution function plot,^{66,67} in which the displacement of the ion pair is plotted against its lifetime. This allows us to analyze the translational mobility of two associated ions within their lifetimes. We then monitor how far they travel together before they break up. Using the same criterion as the one used in the calculation of the aggregation time correlation function, $P(t)$, we considered that an ion pair exists if the center-of-mass distance between cations and anions is less or equal than 7.5 Å. Figure 9 displays

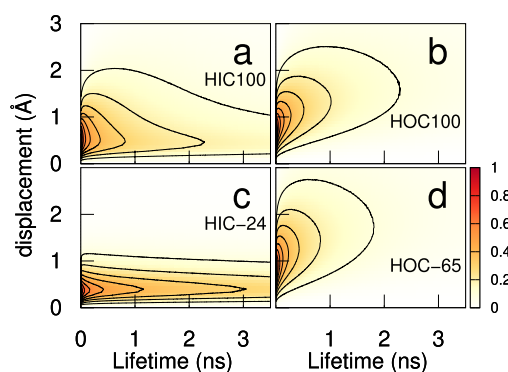


Figure 9. Occurrence density of ion pairs as functions of their lifetimes and displacements. The density is accumulated over ~ 80 ns at 300 K. The plot is normalized with the maximum value of the probability. Top and bottom panels correspond to fully filled pores and monolayer cases, respectively; left and right ones, to HIC and HOC cavities, respectively.

the combined distribution and shows that the majority of ion pairs has a lifetime of the order of a few nanoseconds, consistently with the ion-pair lifetimes listed in Table 4. Note, however, that during that period of time, they are only able to translate a distance of ~ 3 Å or less, a smaller length than the ion–ion distance. Therefore, for most ion pairs, no displacement is achieved before they separate. This fact would suggest that the reduction in the conductivity is likely to be a consequence of two effects: on the one hand, the overall reduction in the mobility of ions, a reduction enhanced by confinement and adsorption phenomena near the pore walls. On the other hand, the IL layering structure induced by the confinement, with the two ionic species within adjacent, coaxial sections, plays a fundamental role in the dynamical cross correlations among the ions. Confinement is responsible not only for the overall slowdown effect but also for a dramatic change in the cross cation/cation, anion/anion, and cation/anion distinct correlations, driving the average motion of these pairs from anticorrelated—in the bulk phase—to positively correlated within the silica pore. These modifications are reflected in the change of sign of the distinct contributions to σ , along with a marked increment in the relative magnitude of the amplitudes of distinct terms, when comparing them with those found in the bulk phase.

4. CONCLUSIONS

In this work, we have studied an imidazolium-like IL [$C_6\text{mimNTF}_2$] by classical MD simulations. The IL was investigated under confinement within silica nanopores of different surface chemistries. We considered hydrophilic pores, in which the inner surface was covered with polar silanol

groups and hydrophobic-like ones, in which 50% of the hydroxyl groups was replaced by TBS chains.

The structural analysis of the local density distributions revealed the presence of a dense adsorbed layer in hydrophilic and hydrophobic pores, leading to IL densities near the pore surface that are a few times larger than in the bulk phase. The orientational functions of IL cations trapped in the hydrophilic pores indicate that the imidazole rings and alkyl chains in close contact with the pore walls are parallel to the surface, whereas the rest of the cations in the first adsorbed layer exhibit their alkyl chain oriented perpendicular to the surface, pointing toward the interior of the pore. In contrast, in the case of the hydrophobic pores, the cations are flipped around, with their alkyl chains oriented perpendicularly to the wall too, but pointing in the opposite direction, toward the pore surface. This change in the microscopic orientation can be explained in terms of the stronger electrostatic interactions between the imidazole ring and the silanol groups at the interface of the hydrophilic cavity.

The analysis of the transport properties of the confined IL along the axial direction revealed an overall retardation in the average diffusive motions of cations and anions in both HOC and HIC pores, compared to their mobility in the bulk phase. The reduction of the diffusion coefficients was found smaller in HOC pores than in HIC ones. In going from a fully filled pore to the lowest simulated loading in HIC systems, the diffusion coefficients exhibited a reduction of one order of magnitude. In contrast, in HOC pores, the silanization of the silica walls significantly reduced this retardation, leading to an enhancement of the overall dynamics. The diffusion coefficients obtained in HOC cavities were roughly independent of filling fraction and remained close to the bulk value. From a *local* perspective, the diffusion exhibited a sharp drop—of one order of magnitude—near the pore walls in hydrophilic cavities, compared to that at the center of the pore, because of hydrogen bonding between the IL and the interfacial SiOH groups. In contrast, in HOC cavities, this effect was less marked because of weaker liquid/wall surface interactions.

The ionic conductivities computed for this IL exhibited features similar to those found for the self-diffusion coefficients. At full loading, the conductivity in the hydrophilic HIC pore was 50% of that in the bulk phase, whereas the functionalization of the pore walls with TBS groups leads to an enhancement of the conductivity, leading to a value of $\sigma_{\text{HOC}} = 0.65\sigma_{\text{bulk}}$ in agreement with experiments.²⁵ Although the absolute values of the ionic conductivities predicted from our simulations resulted underestimated by a factor of ~ 2 with respect to the experimental data, a very good agreement was found between simulations and experiments in the ratio between the conductivity of the trapped IL in HOC pores and the bulk-phase conductivity. The ionic conductivities in HIC pores became gradually reduced with smaller loadings, whereas in HOC cavities, they remained roughly constant and independent of the filling fraction. These trends in the ionic conductivity also agree with those reported in the experimental work of Garaga et al.²⁵ The calculation of local ionic conductivities of the trapped liquids along the radial direction revealed that the conductivity was nonuniform within the silica cavities. Closely related to the ionic mobility, the conductivity resulted constant and close to the bulk value within an inner central region ($0 \leq r \leq 20 \text{ \AA}$) for the HIC pore, followed by an intermediate region ($20 \text{ \AA} \lesssim r \lesssim 35 \text{ \AA}$) where the conductivity was reduced in $\sim 50\%$. Finally, within the adsorbed layer, the

conductivity was 10 times smaller than that at the center of the pore. In contrast, in HOC pores, the variations in the conductivity near the pore walls resulted much weaker. These features suggest that the average values of transport properties contain ponderated contributions from interfacial and bulklike states. When the filling fraction decreases, the relative population of the bulklike states decreases and that of the interfacial states increases. Because in HIC pores, interfacial surface-bound states exhibited a much slower dynamics than bulklike ones, a reduction in fluid loading leads to an overall slower dynamics and lower conductivity. In functionalized cavities, the dynamics of interfacial states remains closer to that of bulk states; as a result, diffusion and conductivity become enhanced with respect to those found in HIC systems, and the dependence of these properties on loading is much weaker.

A systematic analysis of the relative weight of the various contributions to the collective conductivity allowed us to conclude that the behavior of the IL under confinement is very different to that in the bulk phase. In the bulk phase, the dominant features of ion pairs of like and unlike charges are anticorrelated, whereas the distinct components associated to those pairs are all positively correlated when the IL is trapped within the silica pores. These cross-dynamical correlations are largely determined by the characteristics of the detailed local densities. We conclude that the layering structure near the interface induced by confinement would be responsible for an enhancement of cooperative effects between distinct ions within the layers.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b11444.

Time dependence of longitudinal MSDs and self-diffusion of bulk and confined ILs (Figures S1 and S2) and cation–anion radial pair correlation (Figure S3) within different pores (PDF)

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Notes

The authors declare no competing financial interest.

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